MILITARY SPECIFICATION

WATER BATH, ELECTRIC, 120/230 Volt, 50/60 Hz AC

This limited coordinated military specification has been prepared by the Defense Personnel Support Center, based on currently available technical information. However, pending its promulgation as a coordinated military specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This specification covers a water bath for use with an impeller type circulating pump for thawing frozen blood.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center ATTN: Directorate of Medical Materiel DPSC-RST, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC/NA FSC 6640

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SPECIFICATIONS

FEDERAL

Military

MIL-D-42048 - Deployable Medical Systems, General Requirements for.

STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

3. REQUIREMENTS

3.1 General. The complete requirements for acquiring the water bath described herein shall consist of this specification and the latest issue of Military Specification MIL-D-42048, except where otherwise indicated.

3.2 Design and construction.

3.2.1 Design. Shall be a nominal 31 gallon (117.2 liter) capacity water bath for use in thawing frozen blood. Shall be suitable for use with an inserted impeller type 10 liters/minute circulating pump (not required to be supplied with subject water bath) to allow a minimum of 8 frozen blood units, each containing 700 milliliters (ml) of solution, to thaw from -80 degrees (C) to +20 degrees (C).

Shall be designed to operate with a water bath temperature of 37 to 42 degrees (C). When the water bath is heated to and stabilized at the required temperature, the action of an inserted impeller type circulating pump (e.g. Fisher Scientific Catalog No. 13-874-80) shall accomplish the following:

A. Stabilize the water bath temperature within two minutes after the circulating pump is energized to circulate the water.

B. Maintaining the stabilized water bath temperature for a period of 20 to 35 minutes to allow a minimum of 8 frozen blood units, each containing 700 milliliter (ml) of solution, to thaw from -80 degrees (C) to +20 degrees (C).

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Shall be capable of accommodating an insertable 10 liter per minute circulating impeller type pump (see 6.3) having a nominal overall height of 13 inches with 9 inches of it being submersible.

3.2.2 Construction. Shall be of double wall construction with the inner walls (water chamber) fabricated of corrosion resisting metal. The water chamber shall be of welded or single piece seamless construction. The water chamber shall be insulated with non-asbestos and non-combustible material.

3.2.1 Water chamber. The minimum dimensions of the water chamber (inches) shall be 36 long x 18 wide x 9 deep. The inner wall of the chamber shall have an extended lip on all sides for supporting an inserted rack for holding the blood units. The lip shall be located approximately 3 inches from the bottom of the water chamber.

3.2.2 Heating element. Shall be supplied with corrosion resisting steel tubular heating elements(s) that is (are) positioned between the lip around the water chamber wall and the bottom of the water chamber. The heating element(s) shall be submerged by a water filled bath to eliminate any hot spots to anything placed in a blood unit bag.

3.2.3 Pilot light. A pilot light shall be provided on the front of the water bath housing to denote when the element(s) is/are energized.

3.2.4 Accessories.

3.2.4.1 Thermometer. Shall be supplied with a liquid-in-glass partial immersion thermometer for monitoring the temperature of a water bath. The thermometer shall be calibrated in 1 degree (C) increments for a range that includes -10 to 110 degrees (C). The thermometer shall be supported in a holder that is secured to the water bath housing.

3.2.4.2 Wire rack. Shall be supplied with a corrosion resisting wire rack for holding the blood units in the water bath. The wire rack shall be dimensioned to be supported by the lip configuration within the water chamber. The wire rack shall also be suitable for supporting an inserted circulating pump in an upright position.

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3.3 Electrical characteristics. Unless otherwise indicated in a contract or purchase order, the electrical and line cord characteristics shall conform to requirements contained in the latest edition of MIL-D-42048.

3.4 Workmanship. The water bath shall be free from defects which detract from their appearance or impair their serviceability.

Unit. Each (EA). One water bath electric, as specified, constitutes one unit.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 First article inspection. First article inspection shall be performed on the water bath when a first article sample is required. This inspection shall include examination and tests.

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4.1.4 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer of supplier, the NSN, the item identification, the name of the component/material, the lot number and the lot size.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table I. The unit of product for sampling purposes shall be one water bath.

TABLE I. Sampling for examination.

	Inspection level	AQL (percent defects)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5

4.2.2 For tests. Sampling for tests shall be conducted as follows:

a. First article. Where first article sample is required, all tests shall be performed on that unit.

b. Acceptance testing. When a first article sample has not been required, one unit shall be subjected to all tests specified herein. If a first article sample has successfully passed all tests, performance testing shall not be required at acceptance.

4.3 Examination. The water bath shall be examined to determine compliance with the requirements of this specification in accordance with the classification of defects in table II.

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Table II. Categories and Defects*

Categories	Defects
Major	
101	Water bath not complete with all component parts.
102	Water bath not free of wobbling or rocking when placed on a flat horizontal surface.
103	Water bath not free of sharp edges or burrs accessible to an operator during normal use or cleaning.
104	Components not usable or properly secured.
105	Power cord not free of cuts, tears, or abrasion.
106	Power cord not properly secured to cabinet.
107	Blades of power cord not properly aligned.
Minor	
201	Water bath not supplied in a clean condition.
202	Water bath not free of clusters of chips or scratches.
203	Electric control markings not complete, legible, or permanent.
204	Water bath not supplied with required service data that is complete and legible.

*Examination shall not be limited to the classified possible defects listed herein.

4.4 Tests. Tests shall be conducted to determine compliance with this specification.

4.4.1 Operational tests. All water baths shall be tested for compliance with performance requirements. Each water bath shall be connected to a suitable power source, in accordance with the manufacturer's instructions, and operated to demonstrate the proper functioning of mechanical and electrical components. Any evidence of or tendency to failure of any component shall be cause for rejection.

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4.4.2 Temperature control. The water bath and thermometer shall be set up in accordance with manufacturer's instructions. The water bath chamber shall be filled with water at a temperature of 30 degrees (C) or less. The water bath unit shall be connected to a power source and the temperature control shall be adjusted for a bath temperature within a range that includes 37 to 42 degrees (C). After the selected water bath temperature has been reached and stabilized, 10 thermometer readings shall be recorded at 5-minute intervals. A water bath which cannot reach the required temperature range and maintain a stabilized temperature of plus or minus 1 degree (i.e. difference between maximum and minimum thermometer readings) shall be cause for rejection. Sampling for testing shall be conducted in accordance with MIL-STD-105 with an S-1 inspection level and an acceptance number of zero.

5. PACKAGING

5.1 General.

The complete requirements for packaging shall consist of this specification and the latest issue of Military Specification MIL-D-42048 except where otherwise indicated.

6. NOTES

6.1 Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

6.2 Regulatory requirements.

6.2.1 Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

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6.2.2 Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

6.2.3 Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order. Provisioning technical documentation required and appropriate Data Item Descriptions (DID's) should be included as part of the contract.

6.3 Applicable NSN. This specification covers the requirements for the following item:

NSN	Item Identification
6640-01-237-0564	WATER BATH, ELECTRIC, 120/230 Volts, 50/60 Hz, AC

The following stock number covers the circulating pump intended for use with the water bath:

NSN	Item Identification
6640-01-235-1378	PUMP, CIRCULATING, 120/230 Volt, 50/60 Hz, AC

MILITARY INTERESTS:

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review Activities:
Army - MD
Navy - MS
Air Force - 03

User Activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:

DOD-MB

Agent:

DLA-DM

Project No. 6640-1534

Location: Disk No. 14
MIL/SPEC/WATER/BATH/S31

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER
MIL-W-42062(DM)

2. DOCUMENT TITLE
WATER BATH, ELECTRIC, 120/230 Volt, 50/60 Hz AC

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional

b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional

c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional

8. DATE OF SUBMISSION *(YYMMDD)*

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